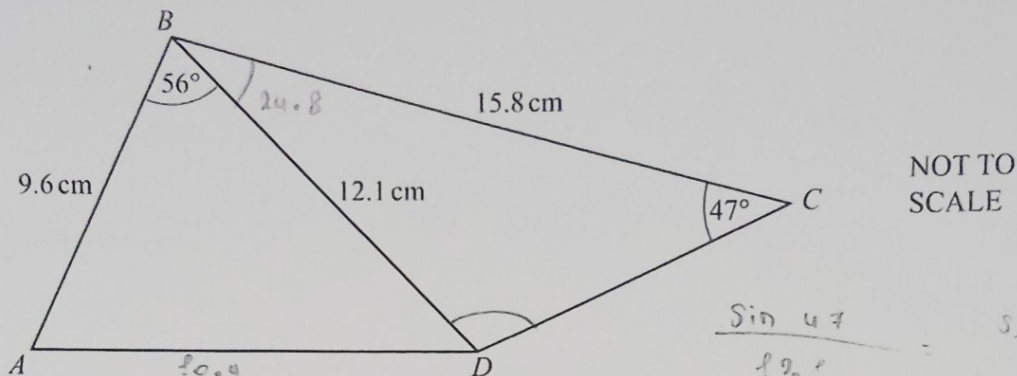


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The diagram shows a quadrilateral $ABCD$.

(a) Calculate AD .

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$c^2 = 9.6^2 + 12.1^2 - 2 \times 9.6 \times 12.1 \cos 56$$

$$c^2 = 238.57 - 199.91$$

$$c^2 = 108.66$$

$$c = \sqrt{108.66}$$

$$c = 10.4 \text{ cm}$$

$$AD = 10.4 \text{ cm [3]}$$

(b) Calculate the obtuse angle BDC .

$$\frac{\sin 47}{12.1} = \frac{\sin \theta}{15.8}$$

$$\sin^{-1}(0.95) = \theta \quad \text{working on page}$$

$$79.8 = \theta$$

$$\frac{\sin 47}{12.1} \times 15.8 = \sin \theta$$

$$180 - 79.8 = 100.2$$

$$0.95 = \sin \theta$$

$$\text{Angle } BDC = 100.2^\circ \text{ [4]}$$

(c) Calculate the area of the quadrilateral.

$$= \frac{1}{2} ab \sin C$$

$$= \frac{1}{2} \times 12.1 \times 9.6 \sin 56$$

$$= 48.2 \text{ cm}^2$$

$$= \frac{1}{2} ab \sin C$$

$$= \frac{1}{2} \times 15.8 \times 12.1 \sin 24.8$$

$$= 41.4 \text{ cm}^2$$

$$89.6 \text{ cm}^2 \text{ [3]}$$

$$= 41.4 + 48.2$$

$$= 89.6 \text{ cm}^2$$

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